# SENTINEL EDGE

Raspberry Pi Edge AI Security Research

**Document purpose:** public-facing technical overview and practical orientation; not a security certification or independent audit.

**Project status:** Architecture planning and research; no project repository verified

**Technology / architecture:** Raspberry Pi 5 (8 GB), AI HAT+ 2 (40 TOPS), Linux, Python, Docker

## Architecture & capabilities

• Proposed local event collection and security telemetry processing

• Research on-device AI inference and monitoring

• Evaluate deployment and performance on resource-constrained hardware

• Possible integration with authorized network or camera data

## Evaluation / getting started

• Prepare Raspberry Pi OS, device updates and accelerator drivers from official sources

• Benchmark a single measurable proof of concept before broad feature claims

• Keep all monitored sources authorized and restrict network exposure

## Limitations and security considerations

• No completed deployed security product is claimed

• Detection quality and supported accelerator workloads are unverified

## Engineering roadmap

• Validate HAT tooling and inference compatibility

• Measure latency, RAM, CPU and power

• Document accuracy and limitations before expanding

## Source and verification

No verified public project repository is available for this concept.

Information reflects the accessible project README and the supplied engineering resume. Features may change; verify current repository branches, releases and documentation before deployment. Do not upload private keys, user data, internal logs or unredacted evidence.